

The Core Challenge

There is a fundamental gap between a customer's real-life aspirations and the complex financial products available today.



Complex Products

Users struggle to map aspirations to technical brochures.



Static Tools

Current sites only offer generic lists or basic calculators.

The Solution: LifeMap Engine

We need a solution that makes financial planning conversational, visual, and highly personalized—transforming "selling a policy" into "planning a life."

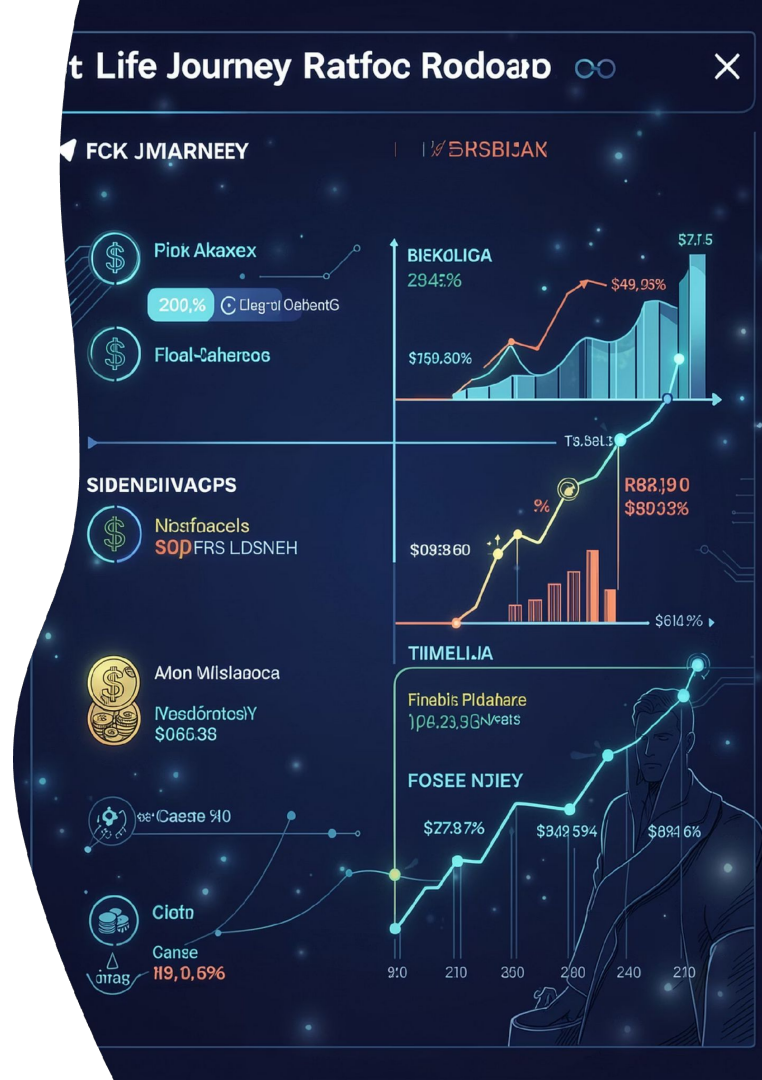


LifeMap

An AI-curated life-goal financial planning platform — replacing static insurance product catalogs with a dynamic, conversational, and deterministic financial journey engine. Built on real ICICI Prudential brochure data.

FINTECH · INSURTECH · AI

TECHNICAL ARCHITECTURE REVIEW



The Problem With Traditional Insurance UX

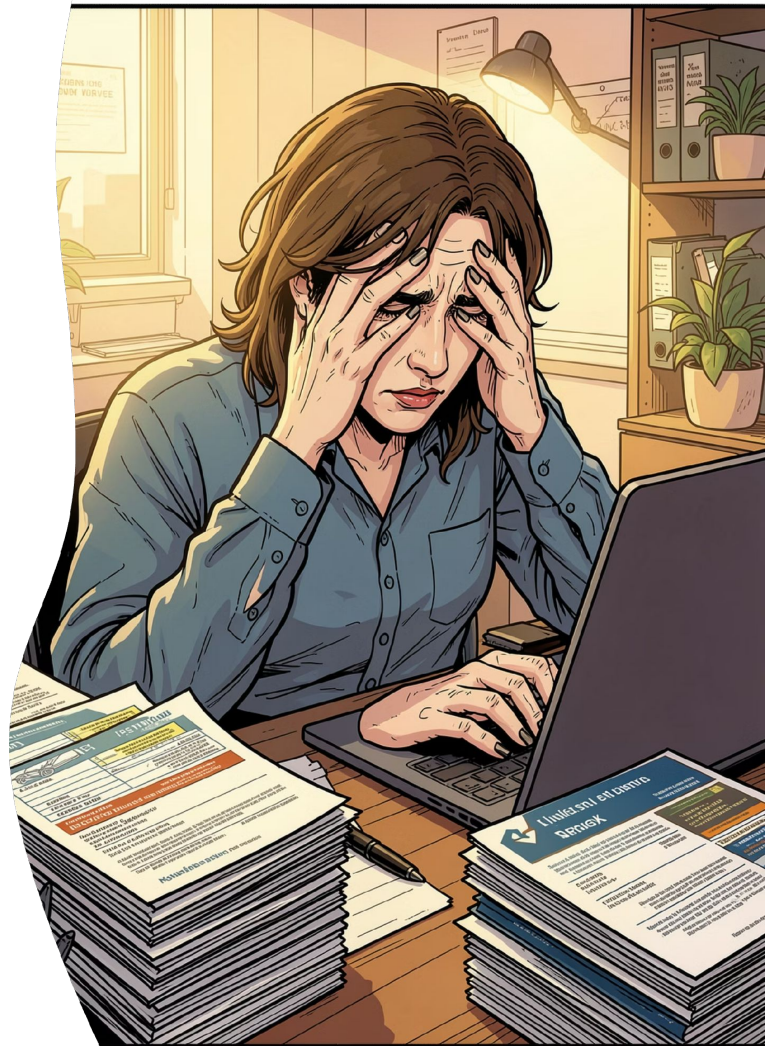
Users don't want product catalogs — they want answers to life questions: **"Can I retire at 55?"** or **"How much for my child's education abroad?"**

Traditional Insurance Site

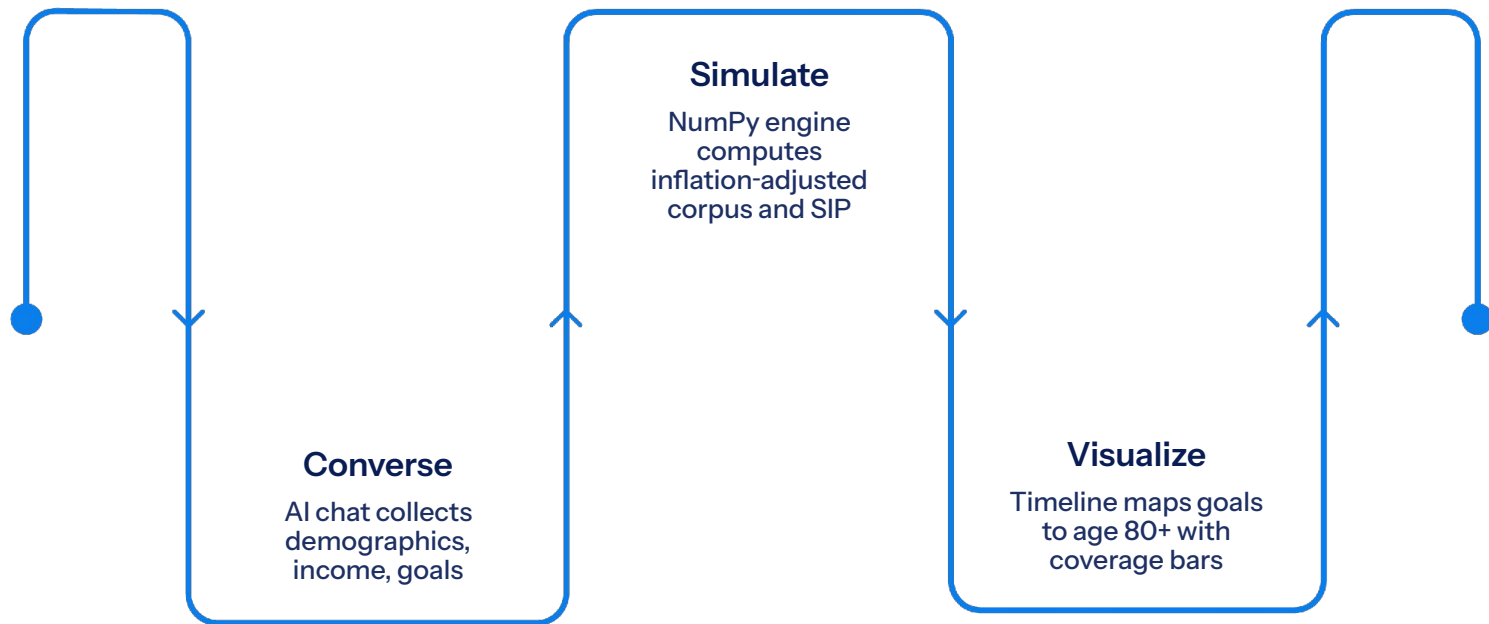
- 47+ products dumped on users
- Static premium calculators
- Generic brochure PDFs
- One-size-fits-all recommendations

LifeMap

- AI-guided life goal discovery
- Dynamic what-if simulator
- Personalized visual Life Journey
- AI-curated, brochure-backed recommendations



Three Steps to a Personalized Financial Journey



The LLM collects context through conversation — the NumPy engine does the math. Every return rate is derived from real ICICI Prudential fund data, not LLM estimates.

NAVIGATION

Dashboard

AI Advisor

Life Journey

Products

Compare

My Report

+ New Chat

RECENT

hi

rams_profile

hi

LifeMap AI Advisor

Online · Powered by GenAI

AI Extract Profile Restart

provide the following missing details?

Financial Goals: Please tell me at least one specific goal (e.g., retirement, child education, buying a home) along with your estimated budget and the timeline/year you want to achieve it

11:21 pm

I've successfully extracted the following information from our chat:

Age: 38**City:** Delhi**Annual Income:** ₹1,200,000**Monthly Expenses:** ₹60,000**Dependents:** 2**Risk Appetite:** Moderate**Goals:** child_education, home_purchase

You can now check your Life Journey timeline.

11:25 pm

Based on your demographics, financials & aspirations

Use Global Profile

Simulate Life Journey

Retirement at 55

Child abroad education

Aggressive ULIP

Safe guaranteed plan

Isolated Chat Variables

IRDAI Compliant

All products are regulated by IRDAI.
Recommendations are illustrative and
not financial advice.

Tell me about your life goals... (Shift+Enter for new line)

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Life Journey

Visualize your goals and financial trajectory over time.

Goal Timeline

Product Simulation

What-If Controls

Reset

Inflation assumption 6%

4% 12%

Annual SIP step-up 8%

0% 20%

Existing savings ₹5.0 L

₹0 ₹50L

Child abroad education

Adds 2.2x to education corpus



Goal Target Ages

Business Fund 55 yrs

46 yrs 85 yrs

Retirement Planning 55 yrs

46 yrs 85 yrs

- Visual Dots:** Each colored dot on the center line represents a life goal. The color tells you the primary recommended product category for that goal (e.g., Orange for Protection, Blue for ULIP).
- Goal Cards:** Shows the total corpus needed at that age, and how much is projected to be covered by the AI's recommended products.
- Coverage Bar:** A quick visual indicator. If the bar is green, you are well covered (80%+). If it's orange, there is a gap you should review.

TODAY (AGE 45)

AGE 55
Business Fund

CORPUS NEEDED

₹17.9 L

₹10.0 L

COST TODAY

Coverage

21%

₹3.8 L covered

Gap: ₹14.1 L

ULIP

Recommended: ICICI Pru Signature

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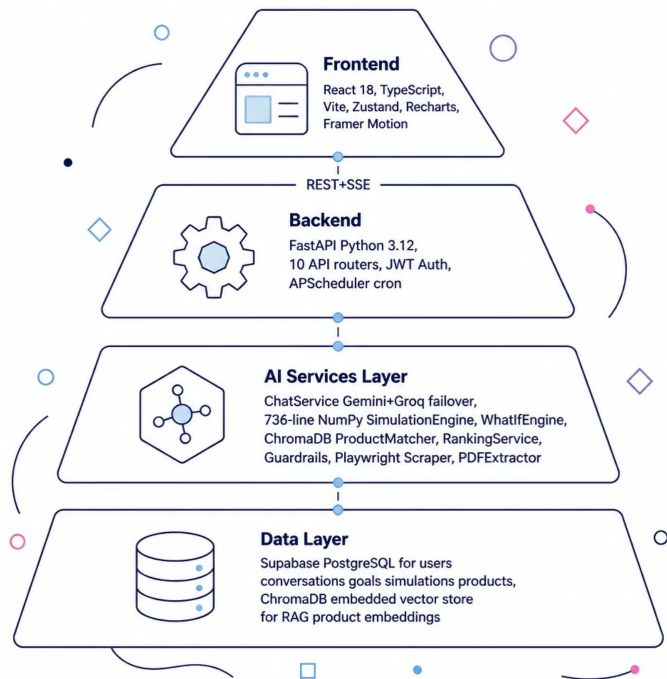
ULIP

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System Architecture

Key Design Decisions

Decision	Rationale
LLM for chat only	Math must be deterministic and verifiable
Dual-LLM fallback	Gemini free tier: 15 RPM → Groq auto-failover
Real brochure data	Returns from ICICI Pru fund data (1.35% FMC)
Vector similarity	ChromaDB embeddings — not keyword rules
SSE streaming	Token-by-token, ChatGPT-like experience



CORE ENGINE

The 736-Line NumPy Simulation Engine

All financial math is handled by NumPy — zero LLM involvement. Returns are derived from real ICICI Pru fund data.



Future Value & SIP

$FV = PV \times (1 + \text{inflation})^{\text{years}}$. Monthly SIP via annuity formula.
Supports stepped SIP with annual increment.



Risk-Adjusted Returns

9-cell matrix (risk $\times$ horizon). Equity ~11% net, Balanced ~9%,
Debt ~7% — all after 1.35% FMC.



ULIP Wealth Booster

Adds 3.25% of average fund value every 5 years from year 10 —
per the IPru Signature brochure.



Gap Analysis

Computes coverage ratio (corpus $\div$ goal cost) and
year-by-year projections for visualization.

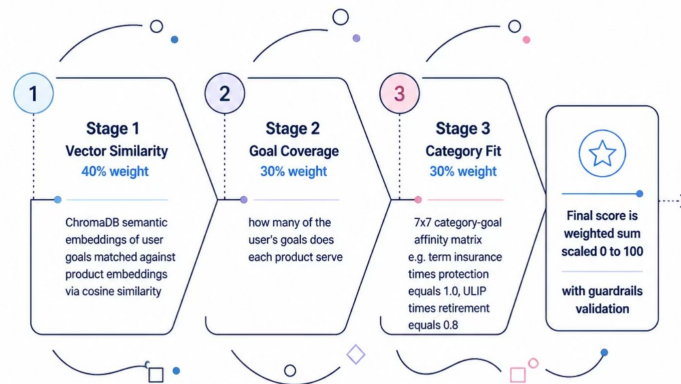
Data Pipeline & Product Matching

Three-Stage Product Matching

From Brochure to Vector Store

5 real ICICI Pru brochures processed end-to-end:

1. **Scrape:** Playwright discovers product pages, downloads PDFs
2. **Extract:** LLM parses first 15 pages → validated JSON
3. **Chunk:** RecursiveCharacterTextSplitter (1K chars, 200 overlap)
4. **Embed:** HuggingFace all-MiniLM-L6-v2 — local, zero cost
5. **Index:** ChromaDB cosine similarity store



Challenges Faced & How They Were Solved

GenAI Rate Limits (Gemini 15 RPM)

Dual-LLM failover: auto-switches to Groq Llama 3.3 70B on 429 errors. 60s cooldown retries Gemini.

LLMs Can't Do Math

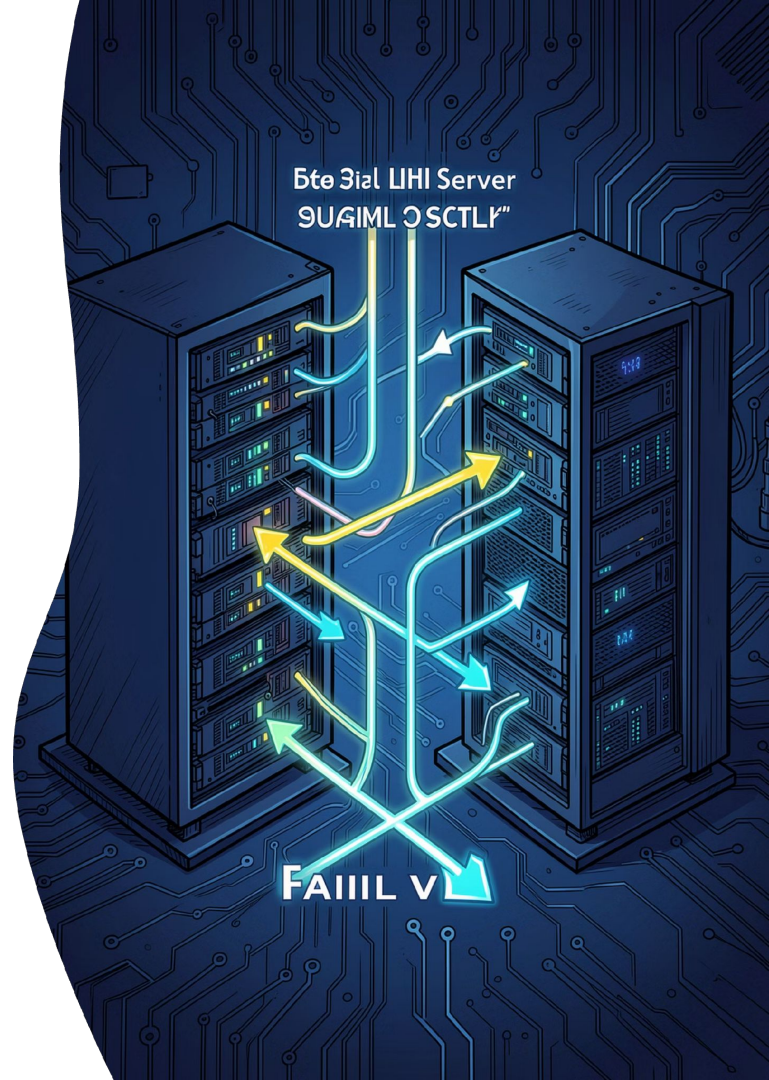
Strict separation: LLM collects context, NumPy handles 100% of calculations. Guardrails clamp SIP to ₹100–₹10L.

Scraping Fragility

fallbacks: local brochure PDFs, and 15s timeout handling. System never goes empty.

Chat Context Isolation

Zustand maintains per-conversation `chatContexts`. Global profile sync available. No cross-chat pollution.



What's Built — 8 Weeks End to End

736

Engine Lines

NumPy simulation engine

5

Brochures Indexed

Real ICICI Pru PDFs

25+

API Endpoints

10 FastAPI routers

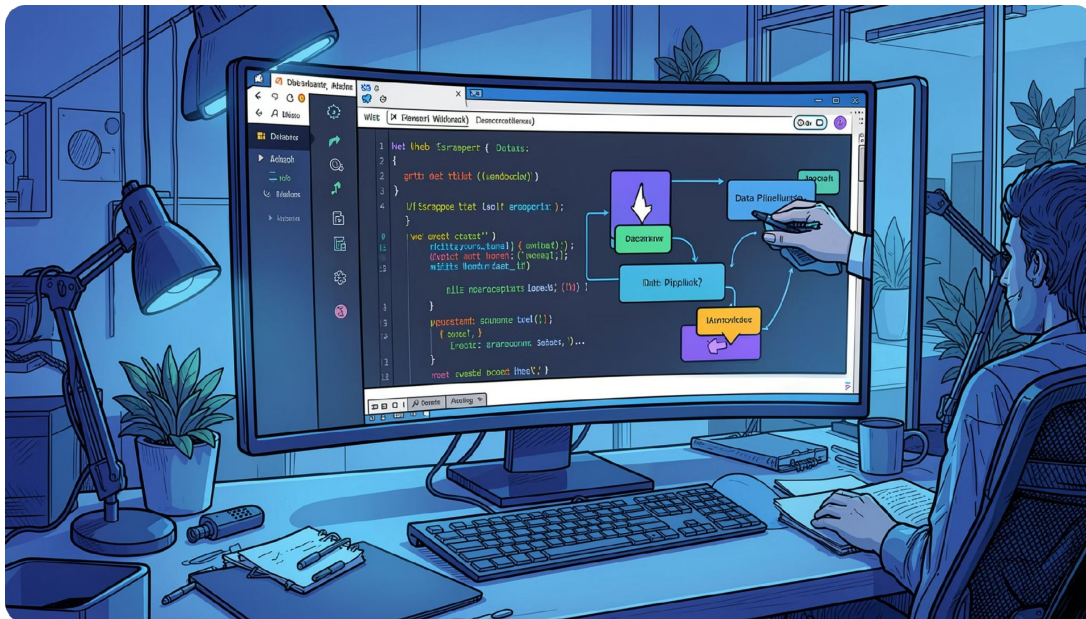
100%

E2E Flow

Chat → Simulate → Visualize

Component	Status	Key Detail
FastAPI Backend	✓ Complete	10 routers, JWT auth, rate limiting
NumPy Simulation Engine	✓ Complete	Brochure-backed returns, Wealth Booster
Dual-LLM Chat + RAG	✓ Complete	Gemini → Groq failover, SSE streaming
What-If Engine	✓ Complete	6 templates, baseline vs. modified delta
Frontend UI	✓ Complete	Chat, Timeline, What-If Panel, Products
Dynamic PDF Fetching	🔄 Next	Playwright scraper ready to wire
PDF Export	✗ Not done	Deprioritized

What's Next



Dynamic PDF Fetching

Playwright scraper is built — navigates ICICI Pru pages, downloads brochures. Currently 5 PDFs pre-loaded for demo stability. Next: wire `scrape_all()` into the live pipeline. Monthly APScheduler cron handles auto-refresh.

Simulation Enhancements

Staggered result card animations, animated coverage bars, smoother Recharts transitions on slider change. Deeper product comparison with projected returns for Product A vs B over the same tenure.

Key Talking Points

Why doesn't the LLM do the math?

LLMs hallucinate numbers. NumPy with brochure-backed rates — every calculation is deterministic and verifiable.

What happens when Gemini hits rate limits?

Auto-switches to Groq Llama 3.3 70B. A 60s cooldown retries Gemini. Users never see an error.

Are return rates realistic?

Derived from ICICI Pru data: Equity ~11% net, Wealth Booster 3.25% every 5 years from year 10.

How are goals matched to products?

Vector search (40%), goal coverage (30%), category-fit matrix (30%) — not keyword matching.



Your Profile

Edit your details so the AI can assist you better



Full Name

john

Age

45

City

delhi

Monthly Income (₹)

80000

Monthly Expenses (₹)

50000

Existing Coverage (₹)

700000

Dependants

2

Risk Appetite

Moderate ▾

Marital Status

Married ▾

Occupation

software engineer

Life Goals

Child's Higher Education

Child's Marriage

Buy a Home

Retirement Planning

Leave a Legacy

Business Fund